Linux (Continue)

Shell Scripting IV: Command sequences & Functions

Overview & Learning Outcomes

This lab you will:

1. Understand how multiple commands are running in a sequencing
2. How to use function

Command Sequences

1. `cmd1 ; cmd2` : execute cmd1, then execute cmd2
2. `(cmd1 ; cmd2)` : In a subshell, execute cmd1, then execute cmd2
3. `{cmd1 ; cmd2}` : In the current shell, execute cmd1, then execute cmd2
4. `cmd1 && cmd2` : execute cmd2 only if cmd1 succeeds
5. `cmd1 || cmd2` : execute cmd2 only if cmd1 fails

Command Sequences / Example

```
touch testmyfile.txt ; mkdir testmydir

(echo "hello world" ; cat hello.txt)

{touch "file1.txt" ; touch file2.txt}

cat not_found_file.txt && echo "hello >.<"

cat not_found_file.txt || echo "hello <3"

echo "I love Linux" || cat not_found_file.txt || echo "hello <3"
```

Function

- Syntax:

```
function_name()
{
    command1
    command2
}

function_name
```

Function
Example 1: Call a function

Example1: Call a function

```
my_show() {
    echo "How are you doing?"
}
echo "Welcome to the kernel!"
my_show
```

Output is:

```
# Welcome to the kernel!
# How are you doing?
```

Function
Example 2: Overwrite global variable

```
my_func() {
    echo "myfunc was called."
    MY_VAR=200
}

echo "myscript was called."
MY_VAR=100
echo "MYVAR before is $MY_VAR"                # MYVAR before is 100
my_func
echo "MYVAR after is $MY_VAR"                 # MYVAR after is 200
```

Function
Example 3: Call a function with one or more arguments

```
my_arguments() {
    echo $1
    echo "$3 / $2"
}



echo "hi there!"
my_arguments "this is" 123 la

# hi there!
# this is
# la / 123
```

```
my_arguments() {
    ARG1=$1
    ARG2=$2
    ARG3=$3
    echo $ARG1
    echo "$ARG3 / $ARG2"
}
echo "hi there!"
my_arguments "this is" 123 la

# hi there!
# this is
# la / 123
```

Function
Example 4: Return

```
my_double_number() {
    ARG1=$1
    RESULT=$((ARG1 * 2))
    echo $RESULT
}
MYNUMBER=$(my_double_number 5)
echo $MYNUMBER

ANOTHER_NUMBER=$(my_double_number "20")
echo $ANOTHER_NUMBER

NUMBER3=`my_double_number "40"`
echo $NUMBER3
```

Function
Example 6: Return on meeting condition

```
my_dd_test() {
    ARG1=$1
    if [[ "$ARG1" -lt "0" ]]; then
        echo 0
        exit
    fi
    RESULT=$((ARG1 * 2))
    echo $RESULT
}
TMP=$(my_dd_test 5)
echo $TMP
```

Function
Example 6: Recursion

```
my_factorial_number() {
    ARG1=$1
    if [[ "$ARG1" -gt "1" ]]; then
        NB_COUNT_DOWN=$((ARG1 - 1))
        TEMP_OF_RECUR_RESULT=$(my_factorial_number "$NB_COUNT_DOWN")
        RESULT=$((ARG1 * TEMP_OF_RECUR_RESULT))
        echo "$RESULT"
    else
        echo "1"
    fi
}
my_factorial_number 5                    # 120
```

Thank You